Quantum Computing: A Layman's Perspective

After Google's dramatic announcement of quantum supremacy, quantum computing has caught the public eye and has generated enormous interest in the subject. Leap, the real-time quantum application environment, even provides one minute of free quantum computing time. This article discusses some aspects of quantum computing from the layman's perspective.

In 2019, Google claimed quantum supremacy and that generated a lot of interest about quantum computing among the general public. But are we on the verge of developing quantum computers that can revolutionise the world? Experts differ on this matter with many claiming quantum computers will be a reality in the very near future, whereas some others (though in the minority) take the extreme opposite view that quantum computers will never be a reality. Whatever might be the eventual outcome, quantum computing is an exciting technology and will remain in the public eye for quite some time.

The quantum computing vocabulary

Let us begin by discussing the two terms that are often associated with quantum computing — quantum superposition and quantum entanglement. I don't have the expertise or audacity to try to explain the above two concepts. Moreover, it takes a lot more than a short article to explain these. But let us, at least, go through the definitions of these two quantum mechanical phenomena to understand why a quantum computer is far more powerful than a classical computer (the computers we use today). According to Wikipedia, a quantum state is a mathematical entity that provides a probability distribution for the outcomes of each possible measurement on a system.

Quantum entanglement is the phenomenon by which the quantum states of a pair of particles are interdependent, irrespective of